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DEVTOOLS / DEBUGGING

SLOW ON MOBILE? STOP GUESSING.

5 CHECKPOINTS TO FIND THE BOTTLENECK

1. **REPRODUCE**
   Network + device
2. **NETWORK**
   Waterfall + bytes
3. **PERFORMANCE**
   Long tasks
4. **AUDIT**
   Lighthouse findings
5. **VERIFY**
   Fix then retest

KEY TAKEAWAY

MEASURE FIRST. CHANGE ONE THING. RETEST.

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DEVTOOLS / NETWORK

WHERE DOES THE TIME GO?

SIZE / WAIT / WATERFALL

REQUEST	WATERFALL
document	
app.js	
hero.webp	
GET /api/items	
analytics.js	

QUESTIONS TO ASK

- Which request has the longest wait?
- Are images or bundles unnecessarily large?
- Is the API responding slowly?

KEY TAKEAWAY

FIND THE SLOW REQUEST BEFORE REWRITING UI CODE.

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DEVTOOLS / PERFORMANCE

IS JAVASCRIPT BLOCKING THE UI?

MAIN THREAD / LONG TASKS / RENDERING

MAIN THREAD (ILLUSTRATIVE)



CHECK THE RECORDING

Long tasks, scripting, layout, painting and user input.

Is a handler doing too much work on one frame?

KEY TAKEAWAY

A SLOW INTERACTION CAN HIDE IN THE MAIN THREAD.

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DEVTOOLS / LIGHTHOUSE

USE THE AUDIT AS A STARTING POINT.

LAB TEST / OPPORTUNITIES / VALIDATION

?	?	?
PERFORMANCE	ACCESSIBILITY	BEST PRACTICES

THE PRACTICAL QUESTION

Can you reproduce the issue in DevTools?

Validate each recommendation before applying it.

KEY TAKEAWAY

ONE LAB SCORE IS NOT EVERY USER EXPERIENCE.

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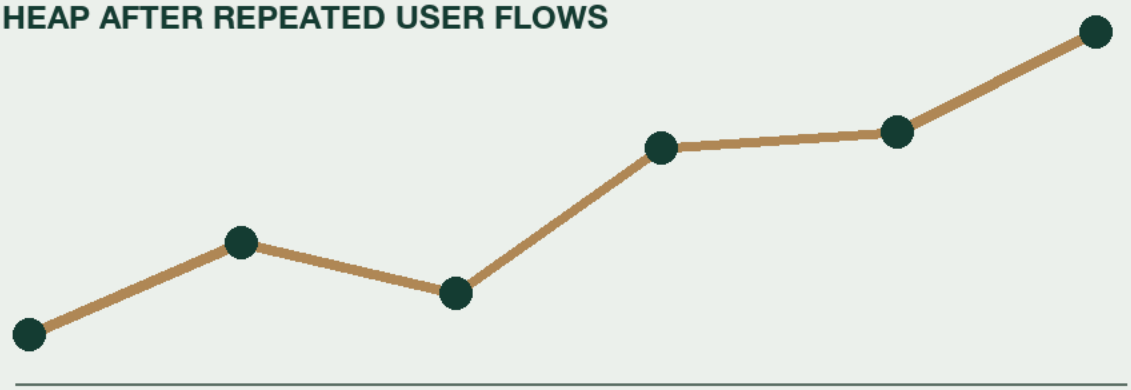
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DEVTOOLS / MEMORY

SLOWER AFTER LOTS OF NAVIGATION?

REPEAT THE FLOW / TAKE MEMORY SNAPSHOTS

HEAP AFTER REPEATED USER FLOWS



CHECK FOR RETAINED REFERENCES

Listeners, timers, cached objects and subscriptions.

Compare snapshots under repeatable conditions.

KEY TAKEAWAY

RISING HEAP ALONE DOES NOT PROVE A LEAK.

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DEVTOOLS / CHECKLIST

THE DEBUGGING PLAYBOOK.

REPRODUCE / MEASURE / RECHECK

- [ ] Reproduce on mobile / throttled network
- [ ] Inspect requests and their waterfall
- [ ] Record a slow interaction
- [ ] Test audit hints, do not trust scores blindly
- [ ] Check memory when slowness grows over time
- [ ] Change one thing, compare before and after

KEY TAKEAWAY

NO OPTIMIZATION WITHOUT A BASELINE.